QBI HACKATHON 2026 · UCSF MISSION BAY · MIT-LICENSED

CryoClear

The open, real-time particle-picking + junk-triage copilot for cryo-EM.

CryoFSL made few-shot picking accurate. **We make picking and junk-cleaning interactive, real-time, and measurable** — with a human-in-the-loop that learns live, on a single GPU.

Computer Science · picking + junk ML + UI

Hardware / Systems · GPU, MRC I/O, streaming

Chemistry · scientific story + HITL

THE GAP

Auto-pickers over-pick junk. Cleaning it is manual, post-hoc, and slow.

- Topaz, crYOLO, CryoSegNet find particles well — but ship no junk-triage layer for ice, carbon edges, and aggregates.
- Scientists catch junk by eye, after the run — discard, re-tune, re-run. Hours after collection, not during.
- Real-time platforms (cryoSPARC Live, Warp) are powerful but closed, and use simple pickers with no correctable junk model.

> **The opening:** no open tool ships a **dedicated, correctable junk-triage classifier with a sub-second human-in-the-loop refit**. That product — not a new picking model — is our contribution.

WHAT CRYOCLEAR DOES

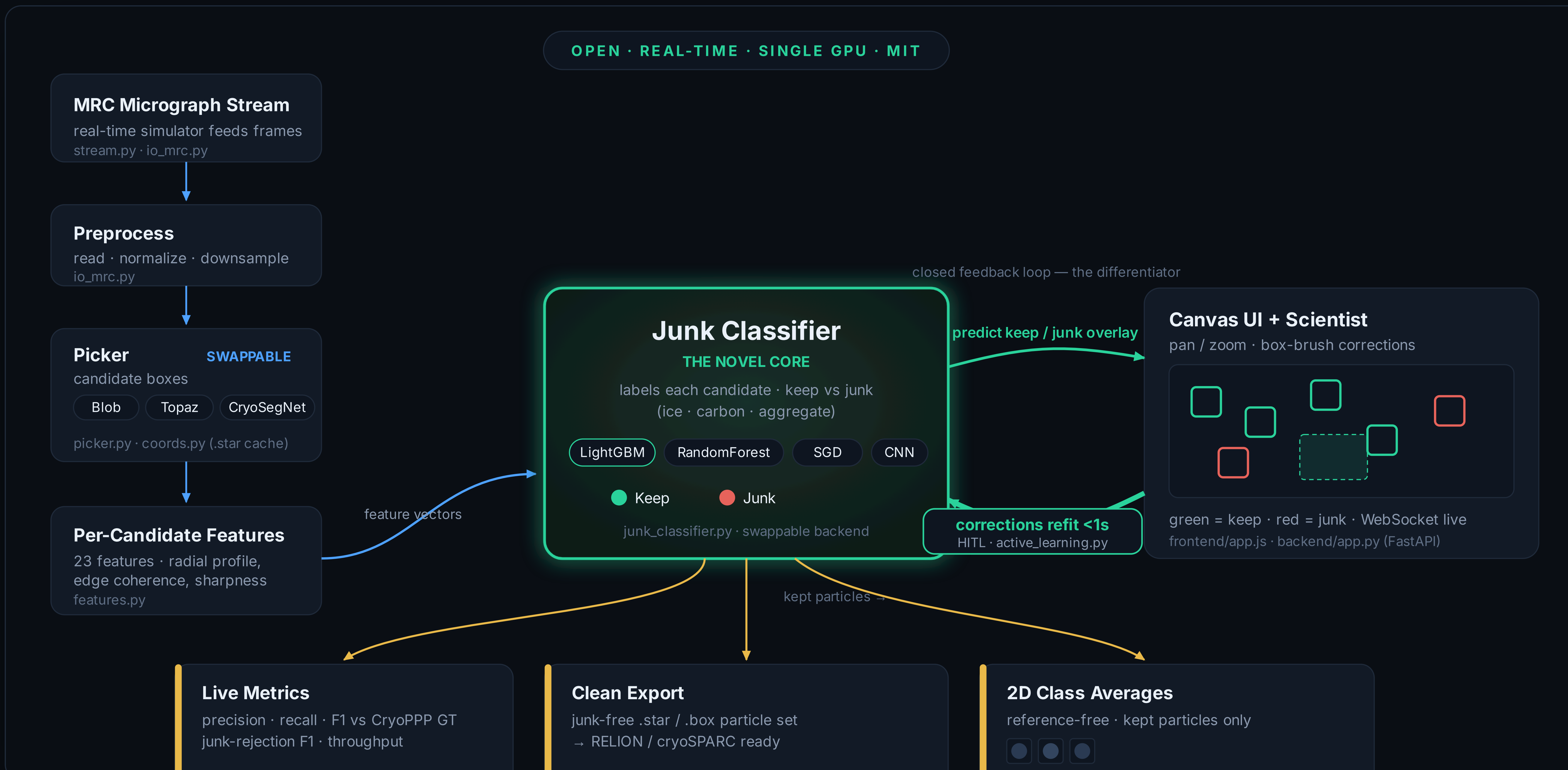
One modular pipeline — swap any picker, triage the junk, learn from the scientist.

MRC stream	Picker	Junk classifier	Your correction	Learns live	Clean export	2D classes
micrographs as they arrive	blob · Topaz · CryoSegNet	keep / ice / carbon / aggregate	box-brush keep/dump · undo	refits in <1s	.star/.box → RELION/cryoSPARC	proof the picks are real

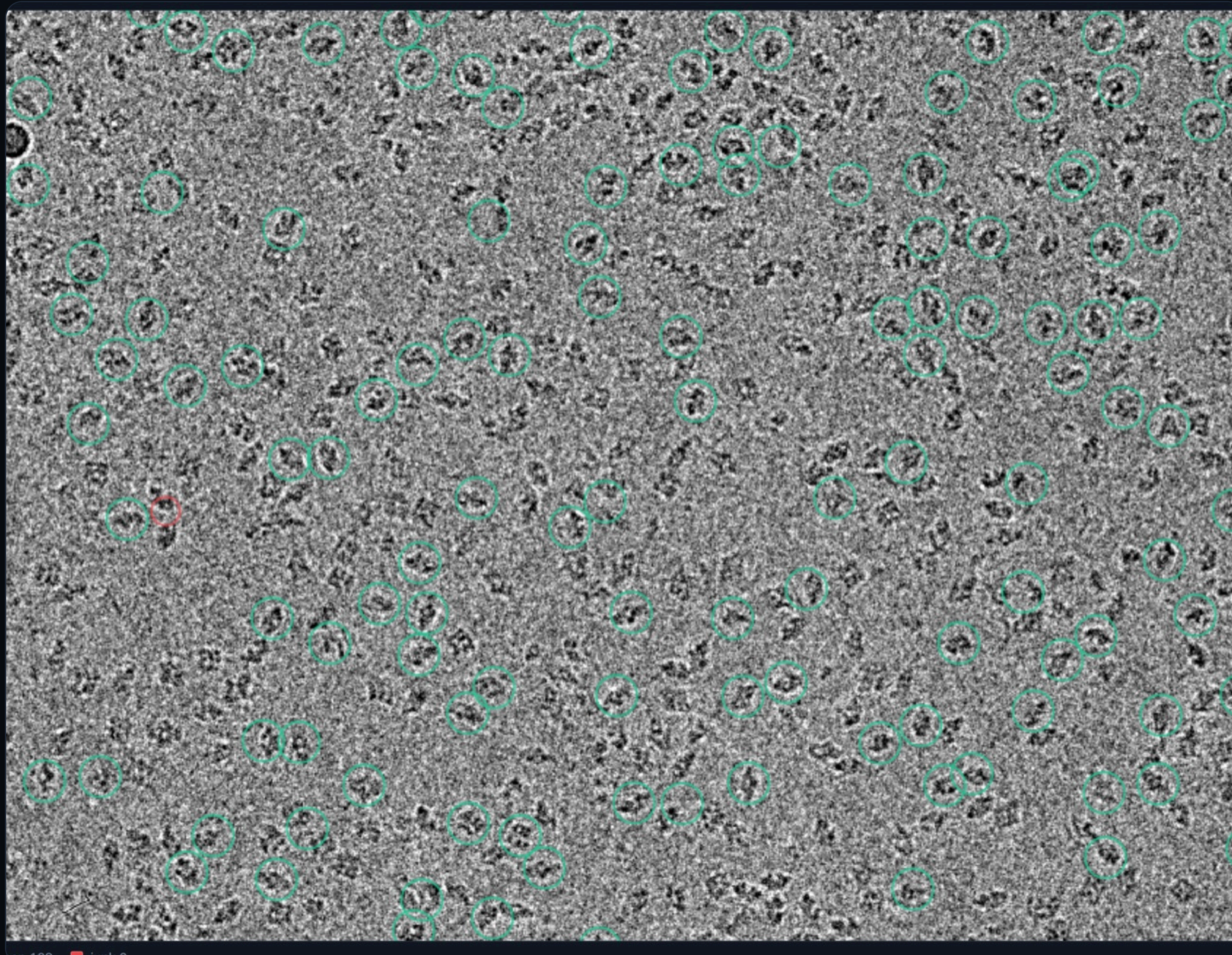
Green = keep, red = junk, overlaid live. Every accept/reject updates the classifier in-session — and a live scoreboard shows exactly how clean the stack is. The default picking + triage path is CPU-light; the GPU is used for the optional deep pickers and the CNN.

SYSTEM ARCHITECTURE

Modular by design — a swappable picker, a novel junk classifier, a live correction loop.



Forward data flow in blue, the human-in-the-loop correction loop in green, measurable outputs in gold. Real module names shown.



EMPIAR-10017 β-galactosidase · each green ring is a kept particle, red is flagged junk.

It picks particles — and tells you which ones to trust.

- Locally-normalised band-pass detector finds candidates across the whole field.
- Each candidate gets 23 intensity-normalised features (radial profile, edge coherence, sharpness).
- The classifier paints keep vs junk in real time.
- Live default is the CPU blob/DoG picker; Topaz & CryoSegNet picks are precomputed and cached as .star, so the app never blocks on the GPU.

MEASURED, AND LABELLED IN-SAMPLE VS HELD-OUT

Junk triage beats raw picks where it counts: a cleaner stack.

+0.09

kept-set purity vs raw picks

in-sample · EMPIAR-10017 · every micrograph we measured

0.47 → 0.64

junk-rejection F1 (LightGBM)

held-out · richer features

27%

junk removed, recall ≈ unchanged

in-sample · at the demo threshold

0.37 → 0.38

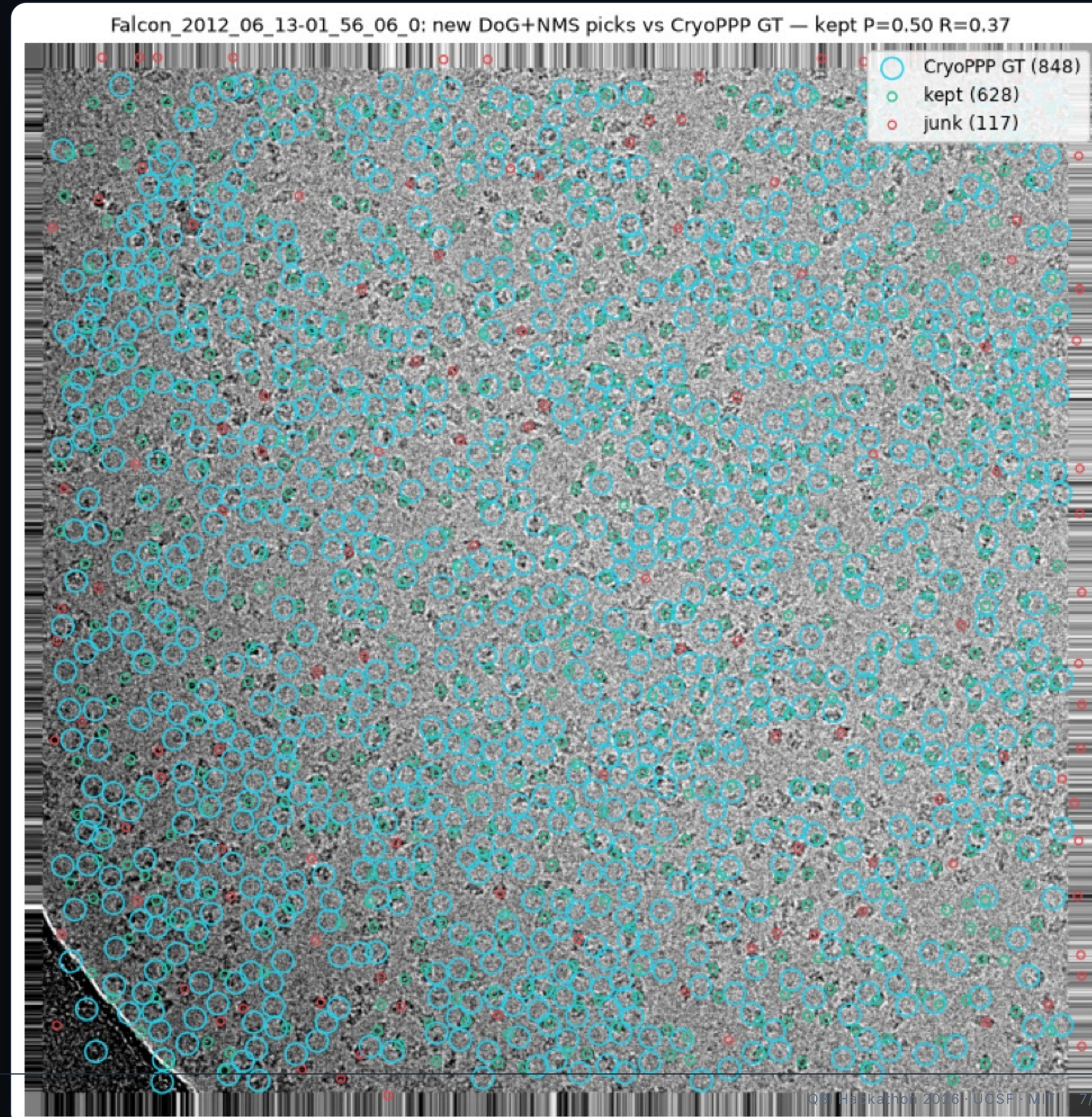
picking F1 vs CryoPPP

held-out · the honest floor

> **Honest ceiling:** on β-gal, picking-F1 after triage ≈ raw — the residual false positives are background that genuinely *resembles* particles, which is what **2D classification, not the picker**, is built to resolve. The classifier's measurable value is purity and a clean kept set, and every number here is tagged in-sample or held-out — no blurring.

Scored against expert CryoPPP ground truth.

- Picks matched to CryoPPP's hand-labelled particles within the particle radius.
- We report precision / recall / F1 per micrograph — held-out, not just in-sample.
- Published deep-learning pickers reach ~0.73 F1 on their own protocols (different data & splits) — shown only as greyed context, never as a head-to-head against our number.

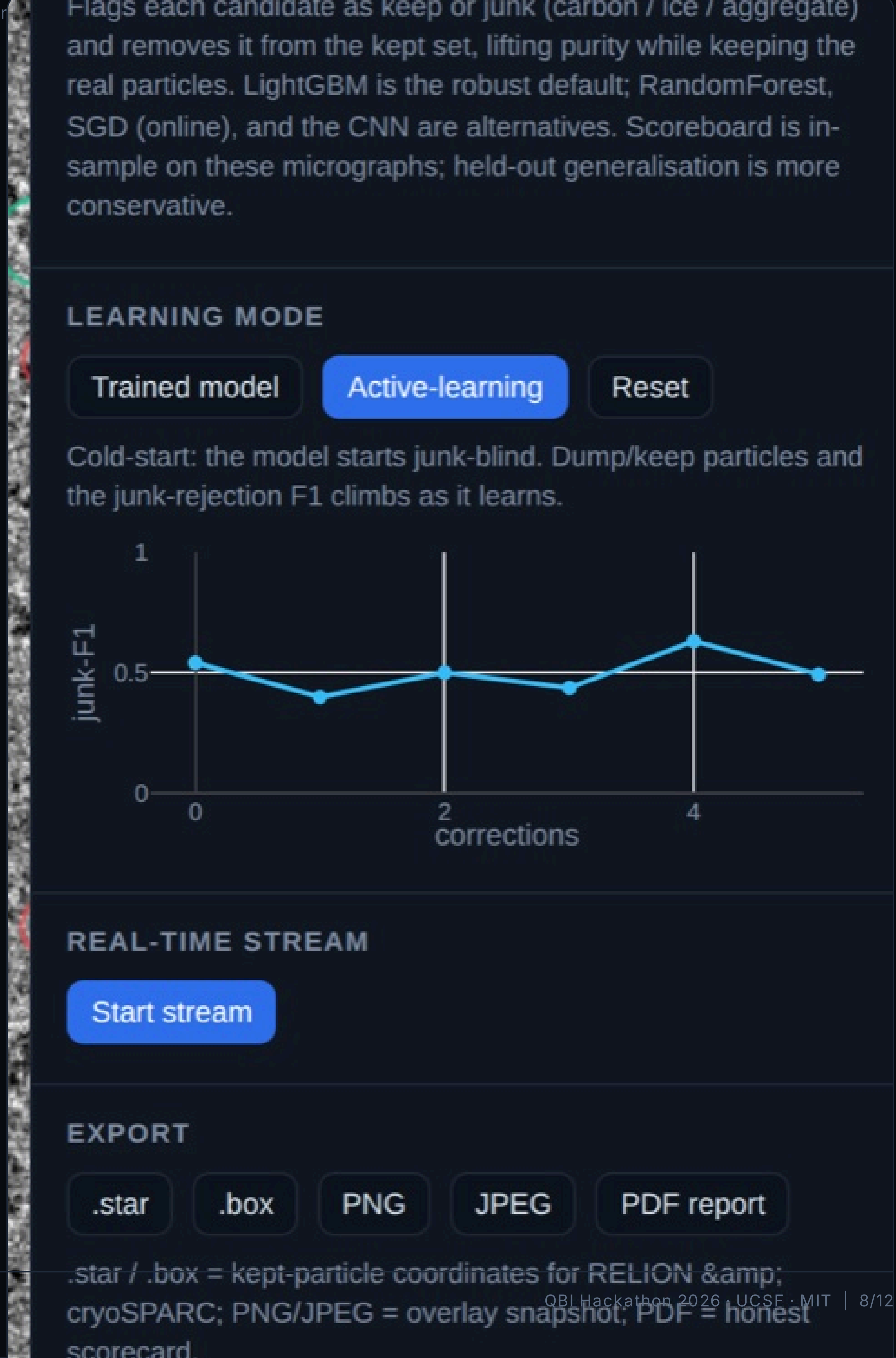


Candidate picks overlaid against CryoPPP gr

It learns from your corrections — live.

- Cold-start: the model begins junk-blind.
- Box-brush to keep or dump particles in bulk; the junk-rejection F1 climbs as it learns.
- Full undo / redo (Ctrl/Cmd+Z); every change flows into the export.

This is the "it learns from me" loop that batch CLI pickers don't have.



Junk-rejection F1 climbing over successive corrections, with one-click exports.

A PLATFORM, NOT A ONE-OFF

The live app: canvas viewer, scoreboard, swappable picker + classifier, multi-dataset switcher.

Everything is swappable and cached.

PICKERS

Blob LoG · CPU
Topaz · GPU
CryoSegNet · GPU

JUNK CLASSIFIERS

LightGBM
RandomForest
SGD · online
CNN

DATASETS

β-gal
TMV
proteasome
+ upload your own MRC

EXPORT

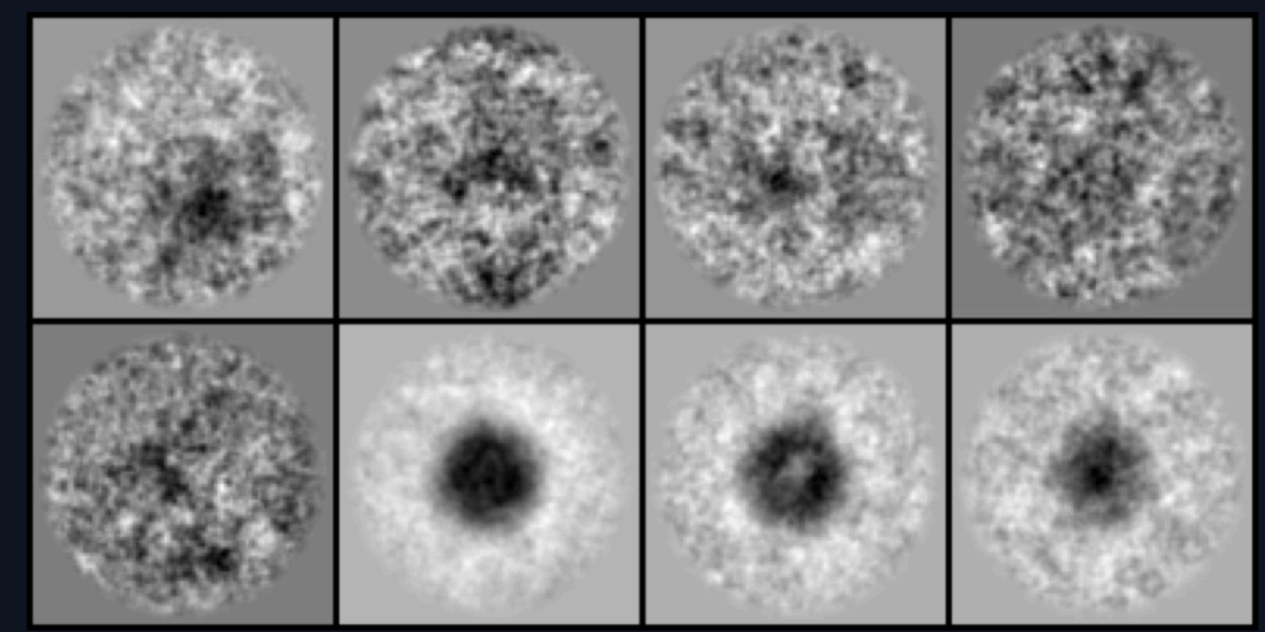
.star
.box
PNG
PDF report

THE PROOF · M4

2D class averages — "these picks are real."

- Reference-free 2D classification of the kept particles.
- Iterative rotational + translational alignment, then k-means clustering on band-pass crops (no CTF, no ML model).
- Recognisable, centred protein views emerge from the cleaned stack.

> **Honest:** β-gal is small, low-contrast, few-view — averages won't match cryoSPARC's textbook crispness without thousands of particles + CTF. The pipeline is industry-shaped; the protein is the limit.



Class-average montage of kept β-gal particles.

WHY IT SCORES

The novelty is the product — and we're honest about what it isn't.

WE BUILT

- An open, real-time junk-triage copilot with a live HITL learning loop.
- Swappable pickers + classifiers, 7+ datasets, upload-your-own, clean export.
- Honest measurement against CryoPPP — held-out, in-sample labelled as such.

> **Rubric fit:** Impact (cleaner stacks, live) · Execution (a usable product, not a notebook) · Background variety (CS + hardware + chemistry) · Novelty (the correctable, sub-second-refit junk-triage UX nobody open-ships).

WE DO NOT CLAIM

- A novel picking model or a novel ML algorithm.
- To beat deep-learning pickers' picking-F1 — swap one in; it's in the menu.
- cryoSPARC-grade 2D on a hard protein from a laptop.

OPEN SOURCE · MIT · SINGLE GPU

Picking + junk-cleaning, interactive, real-time, and measurable.

CryoClear turns post-hoc, manual junk removal into a live copilot that cleans the stack as data streams in — and proves it with honest metrics and 2D class averages.

- Single GPU — core path is CPU-light
- MIT-licensed, fully open
- CS · Hardware · Chemistry